VPC Peering

1. What VPC Peering Actually Is

VPC Peering is a **network connection between two Virtual Private Clouds (VPCs)** that allows resources in both VPCs to communicate **privately using private IP addresses**.

No internet gateway.
No NAT traversal.
Traffic stays **inside AWS network infrastructure**.

In simple terms:
Two isolated networks become directly connected.

Example:

- **VPC-A:** 10.0.0.0/16
- **VPC-B:** 192.168.0.0/16

After peering, an EC2 instance in VPC-A can directly reach an EC2 in VPC-B.

2. Why VPC Peering Exists

AWS isolates VPC networks by default. Without peering, VPCs **cannot communicate**.

Peering is used when you want:

- Communication between **multiple environments**
- Communication between **different AWS accounts**
- Communication between **different regions**

Common cases:

- Dev VPC ↔ Production VPC
- Application VPC ↔ Database VPC
- Shared services VPC ↔ Multiple project VPCs

3. Types of VPC Peering

Intra-Region VPC Peering

Both VPCs are in the **same AWS region**.

Example
Mumbai region:

```
VPC-A (10.0.0.0/16)
      │
      │ Peering
      │
VPC-B (172.16.0.0/16)
```

Characteristics:

- Low latency
- No cross-region cost
- Simple routing

Inter-Region VPC Peering

VPCs are in **different AWS regions**.

Example:

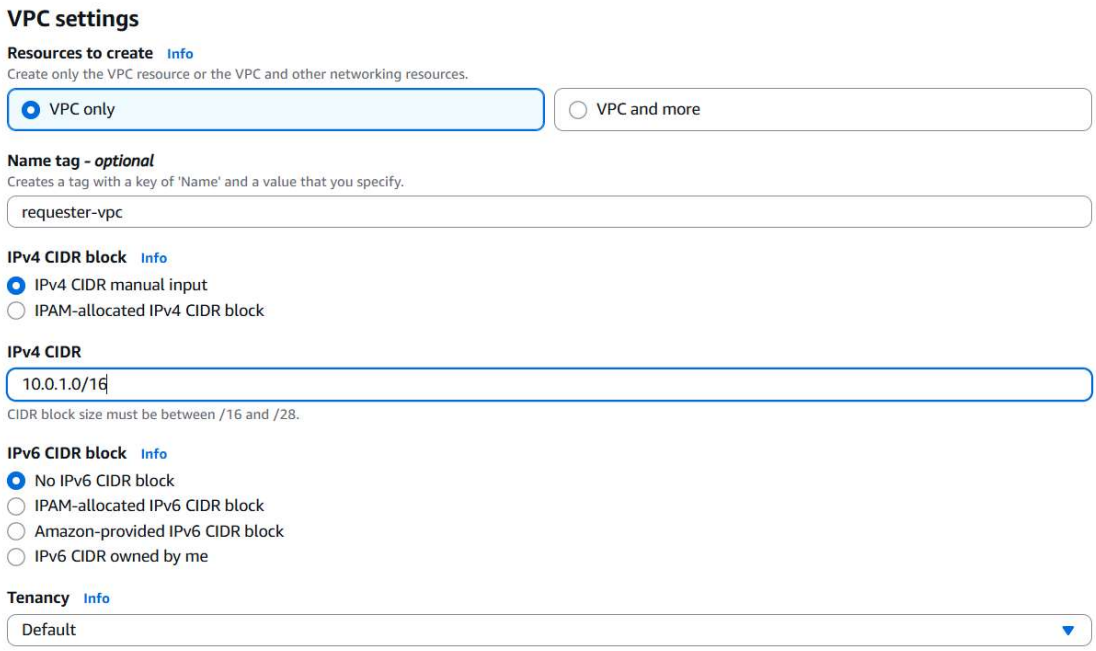
```
VPC-A (Mumbai)
      │
      │ Peering
      │
VPC-B (hyderabad)
```

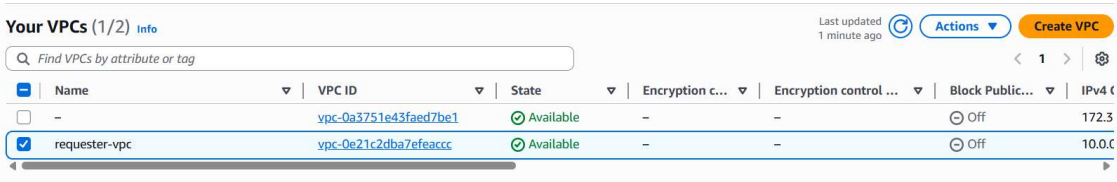
AWS encrypts the traffic automatically.

Used for:

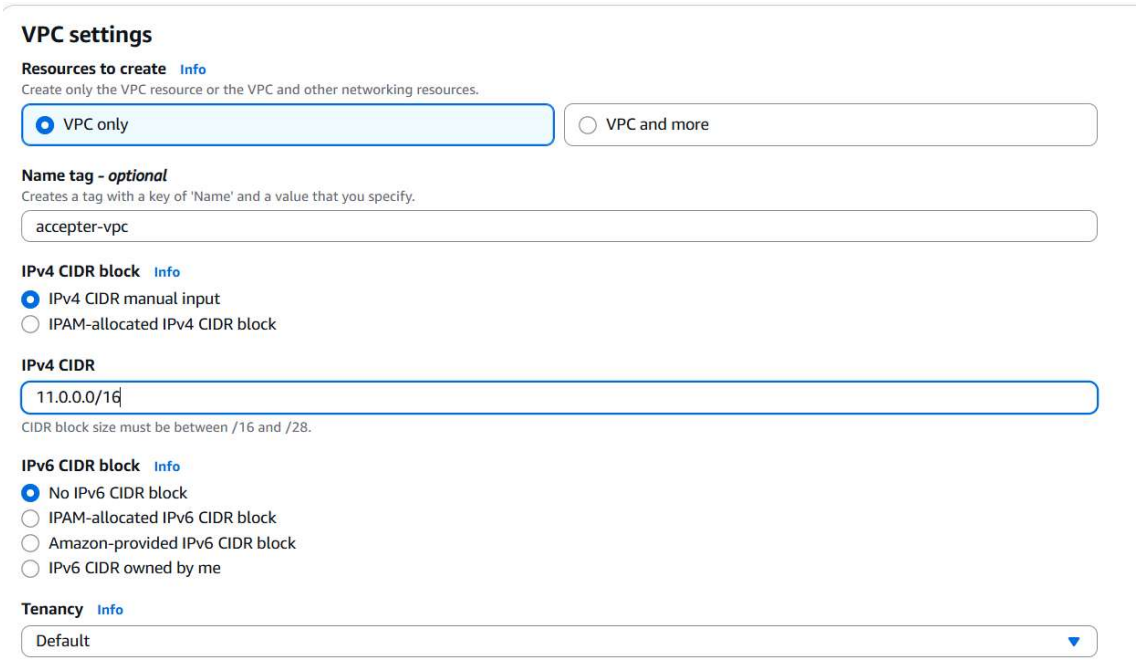
- Multi-region architectures
- Disaster recovery setups

I have created one VPC in the mumbai region.





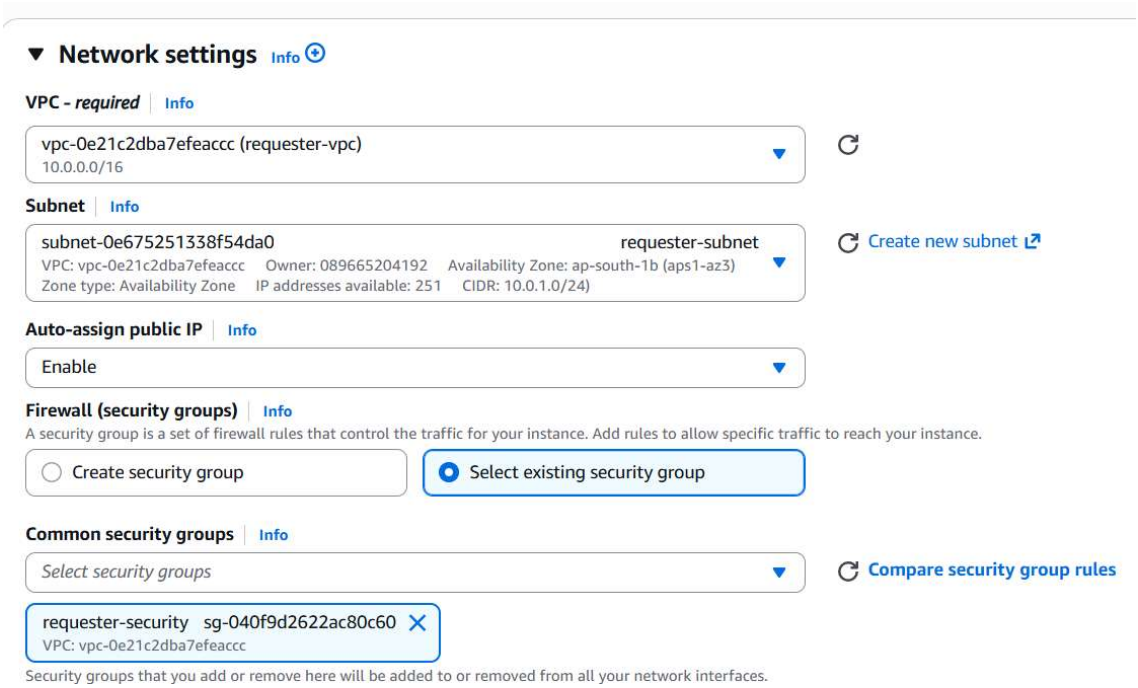
The other in the Hyderabad region.



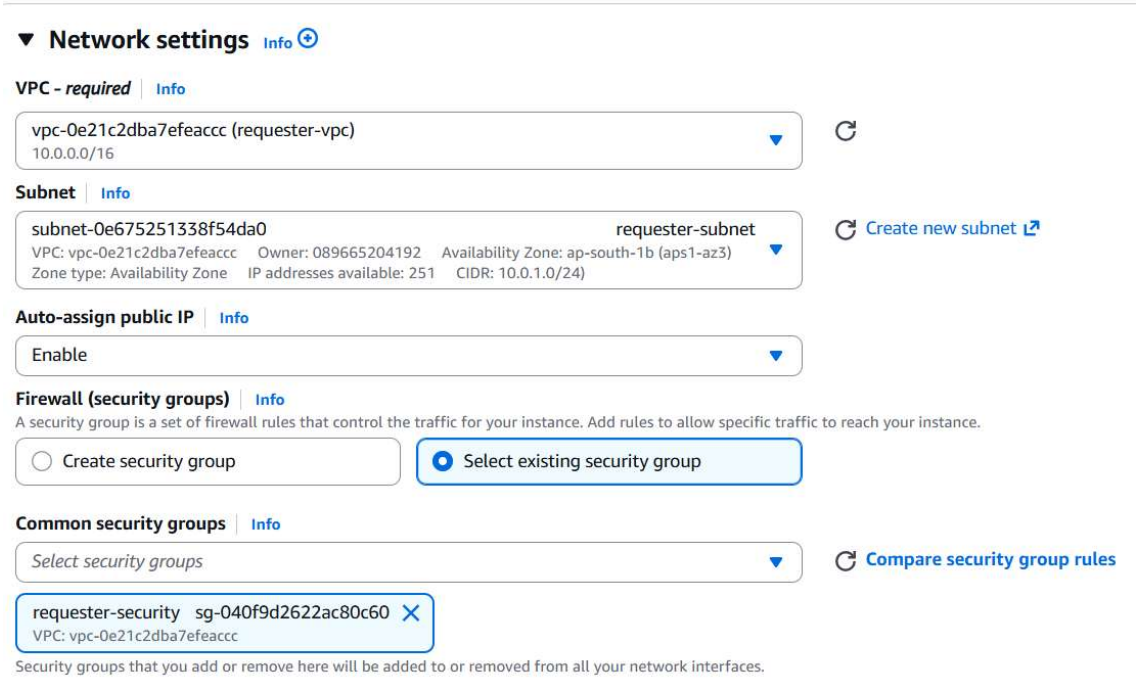
And I have configured all the recommended settings for the VPC to Activate.

(i.e) internet gateway , security groups , subnets.

The creation of public instance in the mumbai region.

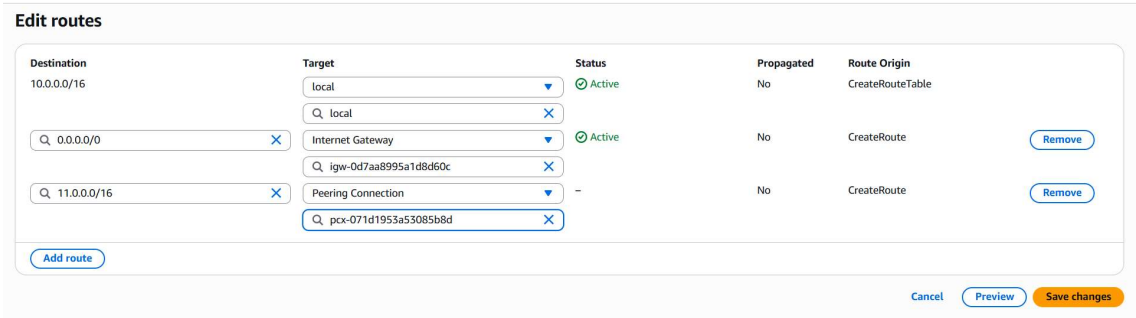


The creation of public instance in the Hyderabad region.

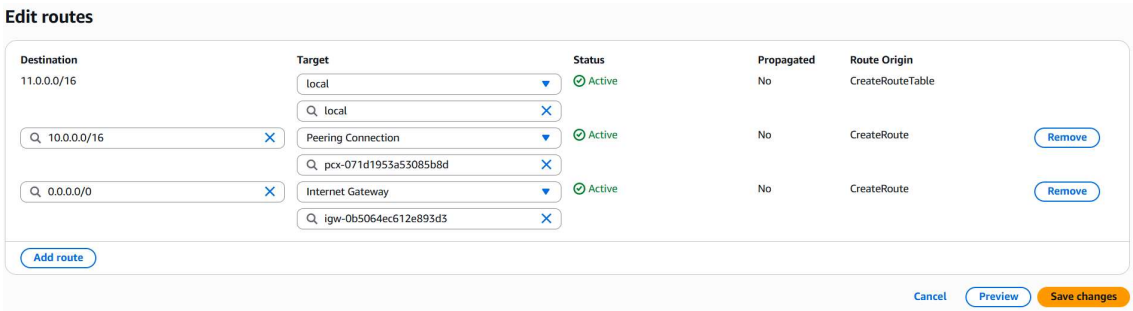


Before peering we have to give one more route values to all the VPC's.

1st VPC.

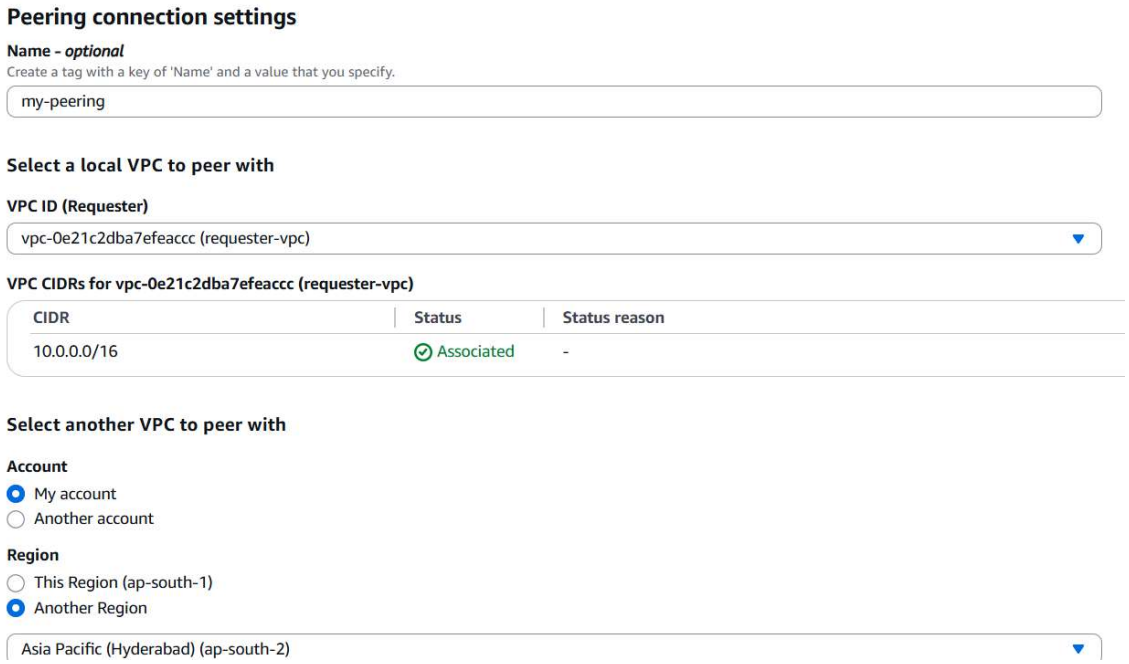


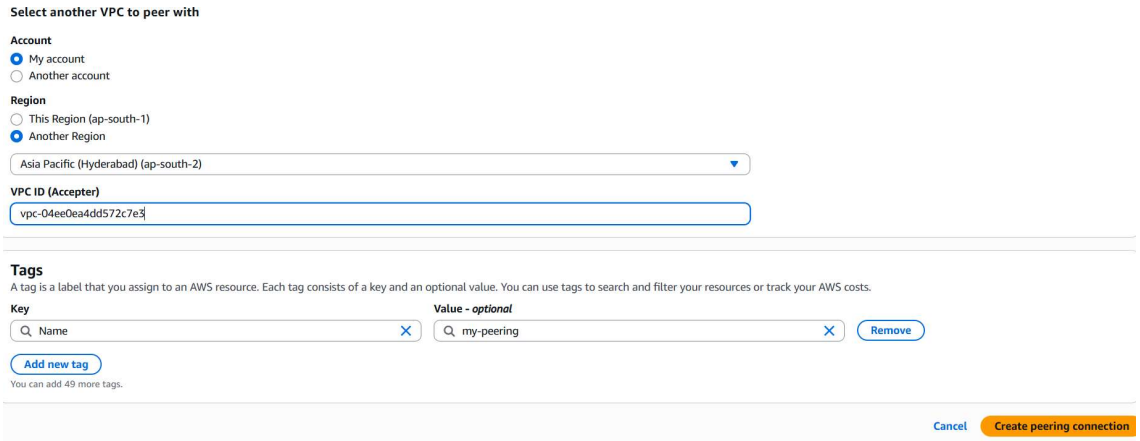
2nd VPC.



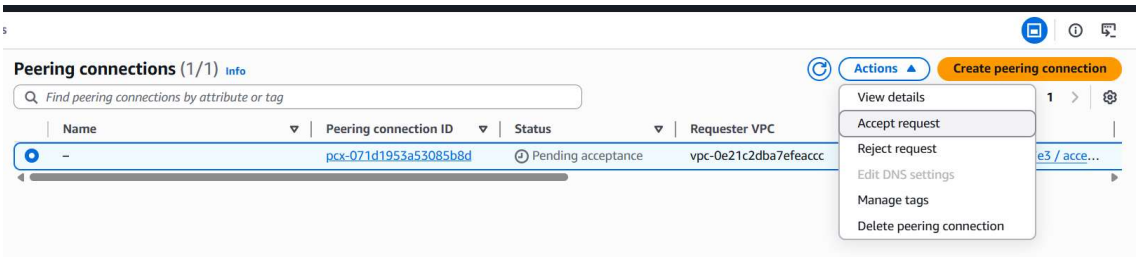
Then this is the Peering of the two VPC's From Different Regions.

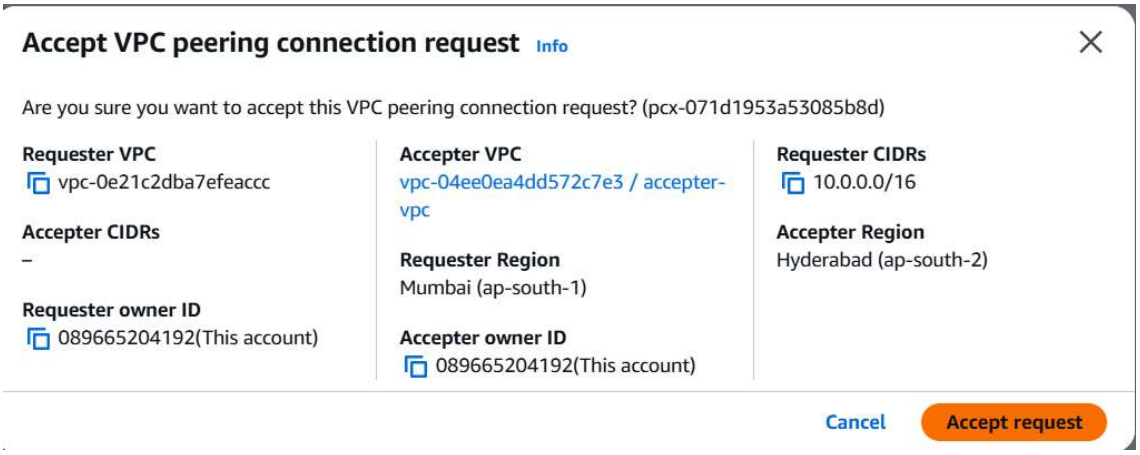
This is the requesting form from the mumbai Region.

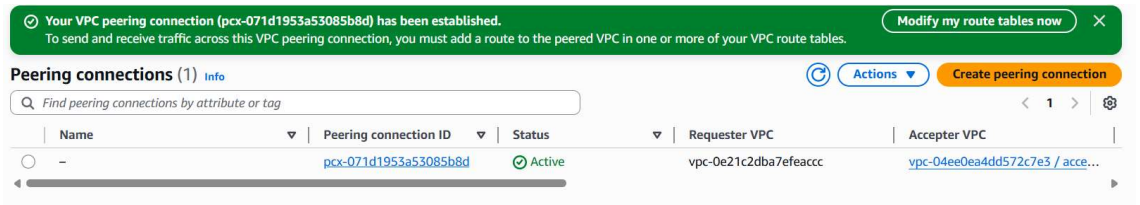




This is the Acceptor form in the Hyderabad Region.







We are using ping command to check whether the internet is connected with the two VPC.

1st Instance.

```
        _/m/'
[ec2-user@ip-11-0-1-141 ~]$ sudo su
[root@ip-11-0-1-141 ec2-user]# ping 10.0.1.246
PING 10.0.1.246 (10.0.1.246) 56(84) bytes of data.
64 bytes from 10.0.1.246: icmp_seq=1 ttl=127 time=14.2 ms
64 bytes from 10.0.1.246: icmp_seq=2 ttl=127 time=14.1 ms
64 bytes from 10.0.1.246: icmp_seq=3 ttl=127 time=14.0 ms
64 bytes from 10.0.1.246: icmp_seq=4 ttl=127 time=14.0 ms
64 bytes from 10.0.1.246: icmp_seq=5 ttl=127 time=14.1 ms
64 bytes from 10.0.1.246: icmp_seq=6 ttl=127 time=14.1 ms
64 bytes from 10.0.1.246: icmp_seq=7 ttl=127 time=14.1 ms
64 bytes from 10.0.1.246: icmp_seq=8 ttl=127 time=14.0 ms
64 bytes from 10.0.1.246: icmp_seq=9 ttl=127 time=14.0 ms
^C
--- 10.0.1.246 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8010ms
rtt min/avg/max/mdev = 14.037/14.077/14.161/0.041 ms
[root@ip-11-0-1-141 ec2-user]# ping 3.108.52.243
PING 3.108.52.243 (3.108.52.243) 56(84) bytes of data.
64 bytes from 3.108.52.243: icmp_seq=1 ttl=120 time=14.1 ms
64 bytes from 3.108.52.243: icmp_seq=2 ttl=120 time=14.1 ms
64 bytes from 3.108.52.243: icmp_seq=3 ttl=120 time=14.1 ms
64 bytes from 3.108.52.243: icmp_seq=4 ttl=120 time=14.1 ms
64 bytes from 3.108.52.243: icmp_seq=5 ttl=120 time=14.1 ms
64 bytes from 3.108.52.243: icmp_seq=6 ttl=120 time=14.1 ms
64 bytes from 3.108.52.243: icmp_seq=7 ttl=120 time=14.1 ms
^C
--- 3.108.52.243 ping statistics ---
```

2nd Instance.

```
aws   Search [Alt+S]
 ~~      \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~      V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-10-0-1-218 ~]$
[ec2-user@ip-10-0-1-218 ~]$
[ec2-user@ip-10-0-1-218 ~]$
[ec2-user@ip-10-0-1-218 ~]$ ping 18.60.184.40
PING 18.60.184.40 (18.60.184.40) 56(84) bytes of data.
64 bytes from 18.60.184.40: icmp_seq=1 ttl=120 time=203 ms
64 bytes from 18.60.184.40: icmp_seq=2 ttl=120 time=203 ms
64 bytes from 18.60.184.40: icmp_seq=3 ttl=120 time=203 ms
64 bytes from 18.60.184.40: icmp_seq=4 ttl=120 time=203 ms
^C
--- 18.60.184.40 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 202.502/202.525/202.573/0.027 ms
```

We are requesting the these two instances public ip with command.